

Supplementary Information

SYNBICT 2.0: a hybrid alignment and exhaustive-search pipeline for exhaustive genetic-part annotation

This document provides the full benchmark dataset, ground-truth construction, per-tool parameter settings, and IGV visualizations of the structural limitation of FlashText longest-match annotation. All values are computed from the run outputs on the 1,166-plasmid Addgene benchmark.

S1. Dataset and feature library

The benchmark comprises 1,166 Addgene plasmids (AddGene2SBOL, sampled set), annotated against a feature library of 4,889 genetic parts (SBOL2 ComponentDefinitions) drawn from nine public collections that contributed parts: Anderson promoters; the CIDAR, Yeast, and Pichia MoClo toolkits; the Voigt terminator and itaconic sets; Cello parts; and a *B. subtilis* collection (bsu, the largest at 3,868 parts). A tenth collection, the EcoFlex MoClo Toolkit, was included but contributed no ComponentDefinitions after import.

Table S1. Feature-library composition by role. A large fraction of terminators, promoters, and RBSs fall below 40 bp, which is where seed-based aligners and BLASTN's significance threshold lose recall. Assembly scars are all 4 bp (MoClo fusion sites) and are excluded by any length threshold ≥ 9 bp.

Role	Count	Min len	Median len	< 40 bp
CDS	2,569	42	741	0
Terminator	1,019	23	43	318 (31%)
Promoter	502	18	53	36 (7%)
RBS	189	9	20	184 (97%)
Engineered region / other	610	1	~30	n/a

S2. Ground-truth construction

The ground truth is built by exact full-length matching, so that any hit a tool reports is genuinely present in the target sequence. Because BLASTN performs exhaustive local alignment, every other tool's exact hits are a subset of the BLASTN hits; false positives are therefore 0 by construction for exact matching at ≥ 14 bp. Below ~ 14 bp, BLASTN's E-value significance threshold suppresses matches at plasmid-length queries, so 9–13 bp parts are recovered by exact substring search instead. The full hybrid ground truth is the union: 2,958 long (≥ 40 bp) + 810 short (14–39 bp) + 174 very short (9–13 bp) parts (174 annotation loci across 173 distinct sequences, one 9 bp part occurring at two loci).

S3. Method comparison for long parts (min_length 40)

Ground truth = BLASTN (word_size 11) = 2,958 parts, all ≥ 40 bp. Time is mean ms per plasmid.

Table S2. Recall and speed on long parts. BLASTN (w11) is the most complete and is competitive in speed; megablast (word_size 28) misses two short terminators. FlashText misses 98 parts, all of which are nested within or overlapping a longer co-located part (95 nested + 3 overlap); these are dropped by its longest-match extraction rather than by a matching failure, and it runs 100 to 1000 times slower. Aligners approach BLASTN's recall only by trading away speed.

Method	Setting	TP	FN	FP	Time (ms)
BLASTN	word_size 28 (megablast)	2,956	2	0	81
BLASTN	word_size 11 (GT)	2,958	0	0	~80
FlashText	exact	2,860	98	0	14,866
BWA	default	2,836	122	0	29
BWA	tuned	2,878	80	0	76
BWA	aggressive	2,949	9	0	384
Minimap2	default	2,655	303	0	59
Minimap2	tuned	2,853	105	0	69
Minimap2	aggressive	2,944	14	0	308

Table S3. Parameter settings added to each base command. Both aligners miss short/nested parts because their defaults drop secondary/short alignments; relaxing that recovers recall with FP still 0.

Tool	default	tuned	aggressive
BWA mem	-a -T 0	+ -k 11 -D 0	+ -W 10
Minimap2	-ax map-ont -N 1 --secondary=no -p 0.9	-ax map-ont -N 50 -p 0.1	-ax map-ont -N 200 -p 0.01 -k 11 -w 3

S4. Short-part regime (min_length 9/14)

Ground truth = BLASTN = 3,768 parts = 2,958 long (≥ 40 bp) + 810 short (14–39 bp). min_length 9 and 14 give the identical BLASTN ground truth.

Table S4. Short-part recall is the dividing line: only aggressive BWA (FN 6) and FlashText (FN 5) recover the 810 short parts; the default/tuned aligners and every Minimap2 setting miss all 810. The "FP" for FlashText (174) and aggressive BWA (17) are real 9–13 bp parts that BLASTN cannot report because of its significance floor (see S5).

Method	Setting	TP	FN 14–39	FN ≥ 40	FN total	FP	Time (ms)
FlashText	exact	3,664	5	99	104	174*	17,446
BWA	default	2,851	810	107	917	0	179
BWA	tuned	2,878	810	80	890	0	321
BWA	aggressive	3,753	6	9	15	17*	697
Minimap2	default	2,687	810	271	1,081	0	105
Minimap2	tuned	2,877	810	81	891	0	107

Minimap2	aggressive	2,895	810	63	873	0	222
----------	------------	-------	-----	----	-----	---	-----

**The FP counts are real 9–13 bp parts (e.g. BBa_B0064_rbs) that BLASTN cannot report at plasmid-length queries.*

S5. Key mechanisms

1. **FP = 0 by construction (≥ 14 bp).** Under exact full-length matching, any hit a tool reports is genuinely present, so BLASTN (exhaustive local alignment) also finds it; every tool's hits are a subset of the BLASTN ground truth. Only relaxing the exact filter introduces FP.
2. **FlashText's limitation is structural, not a matching failure.** Its single-pass, longest-match, non-overlapping scan keeps only the longest part at each locus; a part is dropped whenever it is shorter than, and nested within or overlapping, a longer co-located part. Sequence-identical duplicates at one locus are still both reported; the loss is length-dependent. This is visualized in S6.
3. **BLASTN has a ~ 14 bp reporting floor.** At a plasmid-length query, BLASTN's E-value threshold suppresses matches below ~ 14 bp regardless of word_size, dust, or E-value cutoff (a 12 bp RBS is found when queried alone but not inside a 19 kb plasmid). So BLASTN cannot serve as ground truth below ~ 14 bp; FlashText and aggressive BWA recover such 9–13 bp parts by exact matching.
4. **No single tool dominates across all part lengths.** ≥ 40 bp: BLASTN (w11) is best, being both complete and fast. 14–39 bp: only aggressive BWA and FlashText recover short parts (aligners otherwise miss all 810); FlashText recalls well but is 100 to 1000 times slower. < 14 bp: exact substring search is required. This motivates SYNBICT 2.0's hybrid design.
5. **The speed–recall trade-off is real for the aligners; BLASTN escapes it.** BWA/Minimap2 reach BLASTN-level recall only at 4–13 $\times$ their default runtime, by which point they are slower than BLASTN, which is both complete and fast.

S6. IGV visualization of FlashText's structural misses

Each panel shows one Addgene plasmid with two annotation tracks in IGV: the BLASTN ground-truth track (all exact full-length parts) and the FlashText track. Features that appear in the BLASTN track but not the FlashText track are the parts FlashText drops; in every case a shorter part nested within, or overlapping, a longer co-located part. BED tracks for all examples are provided in igv_export/.

S6.1 addgene155372

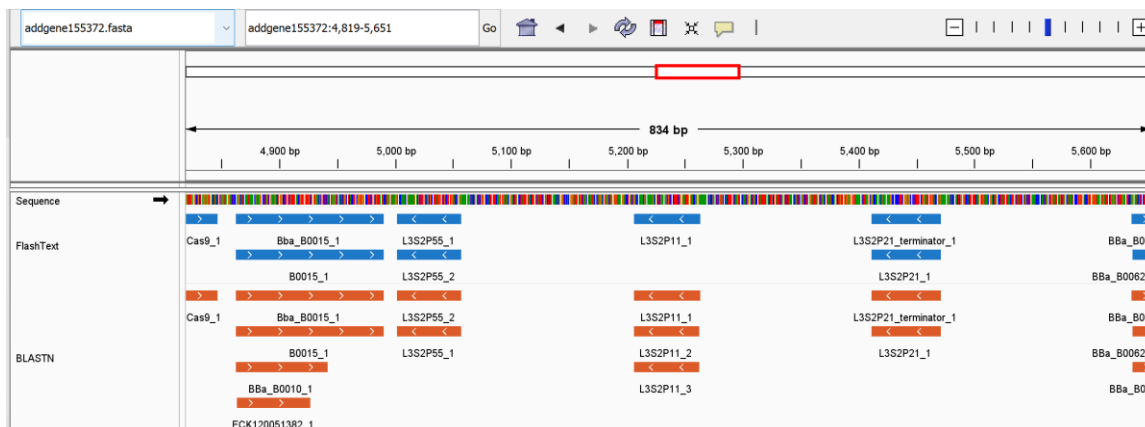


Figure S1. addgene155372: FlashText drops 5 part(s) that BLASTN recovers (BBa_B0010_1, Bba_B0052_1, ECK120051382_1, L3S2P11_2, L3S2P11_3). Each dropped part is nested within or overlaps a longer co-located part, so FlashText's longest-match scan keeps only the longer one.

Part FlashText dropped	Coordinates	Nested within (kept) span
BBa_B0010_1	4861–4941	4861–4990
ECK120051382_1	4862–4926	4861–4990
L3S2P11_3	5205–5262	5205–5263
L3S2P11_2	5205–5262	5205–5263
BBa_B0052_1	5635–5676	5634–5676

S6.2 addgene109150



Figure S2. addgene109150: FlashText drops 3 part(s) that BLASTN recovers (BBa_B0010_1, BBa_B0052_1, ECK120051382_1). Each dropped part is nested within or overlaps a longer co-located part, so FlashText's longest-match scan keeps only the longer one.

Part FlashText dropped	Coordinates	Nested within (kept) span
BBa_B0010_1	794–874	794–923
ECK120051382_1	795–859	794–923
BBa_B0052_1	5194–5235	5194–5235

S6.3 addgene121008

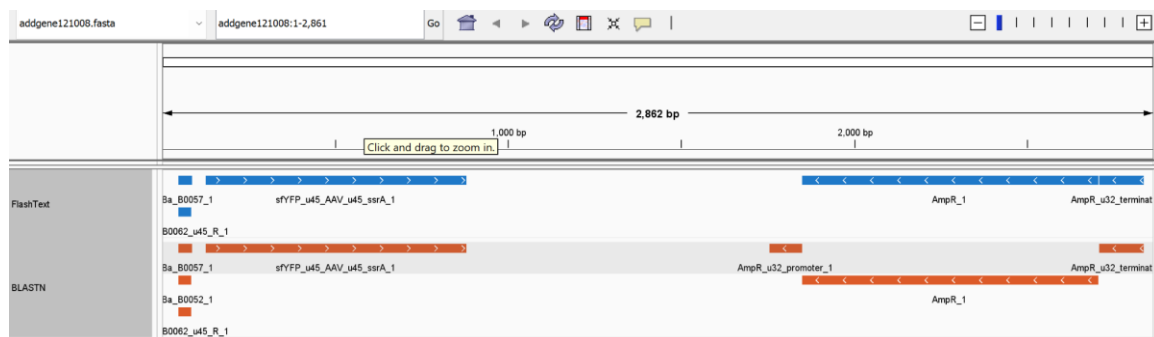


Figure S3. addgene121008: FlashText drops 2 part(s) that BLASTN recovers (AmpR_u32_promoter_1, BBa_B0052_1). Each dropped part is nested within or overlaps a longer co-located part, so FlashText's longest-match scan keeps only the longer one.

Part FlashText dropped	Coordinates	Nested within (kept) span
BBa_B0052_1	44–85	44–85

AmpR_u32_promoter_1	1752–1849	overlapping neighbor
---------------------	-----------	----------------------

S6.4 addgene101703



Figure S4. addgene101703: FlashText drops 1 part(s) that BLASTN recovers (AmpR_u32_promoter_1). Each dropped part is nested within or overlaps a longer co-located part, so FlashText's longest-match scan keeps only the longer one.

Part FlashText dropped	Coordinates	Nested within (kept) span
AmpR_u32_promoter_1	2941–3038	overlapping neighbor

S6.5 addgene107955

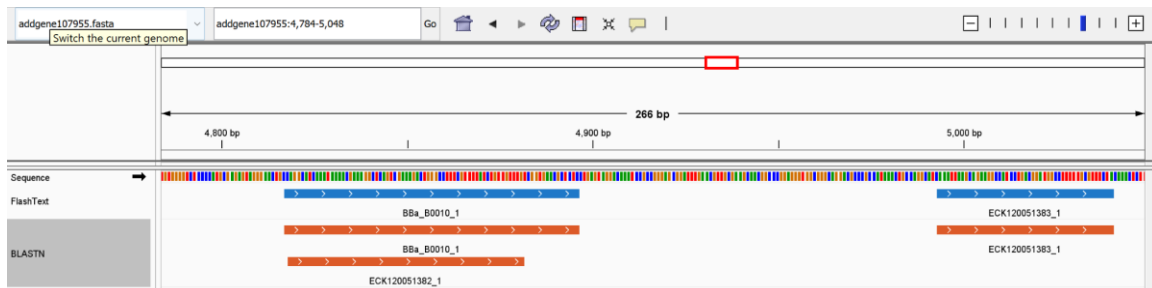


Figure S5. addgene107955: FlashText drops 1 part(s) that BLASTN recovers (ECK120051382_1). Each dropped part is nested within or overlaps a longer co-located part, so FlashText's longest-match scan keeps only the longer one.

Part FlashText dropped	Coordinates	Nested within (kept) span
ECK120051382_1	4817–4881	4816–4896

S6.6 addgene128538



Figure S6. addgene128538: FlashText drops 1 part(s) that BLASTN recovers (aMF_no_EAEA_1). Each dropped part is nested within or overlaps a longer co-located part, so FlashText's longest-match scan keeps only the longer one.

Part FlashText dropped	Coordinates	Nested within (kept) span
aMF_no_EAEA_1	939–1195	939–1206

S6.7 addgene164579



Figure S7. addgene164579: FlashText drops 1 part(s) that BLASTN recovers (ECK120015454_1). Each dropped part is nested within or overlaps a longer co-located part, so FlashText's longest-match scan keeps only the longer one.

Part FlashText dropped	Coordinates	Nested within (kept) span
ECK120015454_1	2705–2750	2697–2752

S6.8 addgene165798



Figure S8. addgene165798: FlashText drops 1 part(s) that BLASTN recovers (AmpR_u32_promoter_1). Each dropped part is nested within or overlaps a longer co-located part, so FlashText's longest-match scan keeps only the longer one.

Part FlashText dropped	Coordinates	Nested within (kept) span
AmpR_u32_promoter_1	864–961	overlapping neighbor

S7. Benchmark plasmid list (Addgene IDs)

All 1166 Addgene plasmids in the benchmark (numerically sorted). Sequences were obtained via AddGene2SBOL.

addgene4	addgene1890	addgene8882	addgene10902	addgene11515	addgene12089
addgene12454	addgene13051	addgene13719	addgene14453	addgene14948	addgene15738
addgene16361	addgene17222	addgene17654	addgene18834	addgene19415	addgene20147
addgene20864	addgene21341	addgene22197	addgene22605	addgene23348	addgene23973
addgene24657	addgene25965	addgene26634	addgene27138	addgene28019	addgene30130
addgene31796	addgene32635	addgene34593	addgene35676	addgene37321	addgene38777
addgene40640	addgene41808	addgene43828	addgene45457	addgene46825	addgene48080
addgene49792	addgene50591	addgene51553	addgene52659	addgene53985	addgene54610
addgene54987	addgene55352	addgene55896	addgene56302	addgene57176	addgene57451
addgene57736	addgene58154	addgene59323	addgene60560	addgene61427	addgene62226
addgene62937	addgene64324	addgene65381	addgene66299	addgene66845	addgene68026
addgene69047	addgene70002	addgene71287	addgene72269	addgene73031	addgene73887
addgene74350	addgene75131	addgene75528	addgene75628	addgene75729	addgene75830
addgene75931	addgene76032	addgene76132	addgene76233	addgene76333	addgene76433
addgene76536	addgene76636	addgene76736	addgene76837	addgene76937	addgene77038
addgene77139	addgene77240	addgene77341	addgene77441	addgene77542	addgene77643
addgene77746	addgene77847	addgene77947	addgene78049	addgene78312	addgene78836
addgene79511	addgene80380	addgene80624	addgene81032	addgene81311	addgene81412
addgene81513	addgene81621	addgene81729	addgene81842	addgene82040	addgene82198
addgene82340	addgene82664	addgene82830	addgene82940	addgene83262	addgene83785
addgene84129	addgene84306	addgene84483	addgene84817	addgene85139	addgene85455
addgene85680	addgene85857	addgene85996	addgene86136	addgene86293	addgene86498
addgene86686	addgene86840	addgene87029	addgene87178	addgene87311	addgene87428
addgene87597	addgene87728	addgene87889	addgene88374	addgene89078	addgene89260
addgene89433	addgene89595	addgene89732	addgene89909	addgene90044	addgene90165
addgene90282	addgene90416	addgene90558	addgene90660	addgene90762	addgene90863
addgene90967	addgene91078	addgene91197	addgene91302	addgene91411	addgene91519
addgene91625	addgene91769	addgene91895	addgene92031	addgene92220	addgene92354
addgene96907	addgene97036	addgene97160	addgene97406	addgene98201	addgene98328
addgene98503	addgene98628	addgene98823	addgene99068	addgene99229	addgene99375
addgene99553	addgene99707	addgene99874	addgene100023	addgene100272	addgene100404
addgene100537	addgene100658	addgene100762	addgene100912	addgene101073	addgene101220

addgene101347	addgene101703	addgene101842	addgene102361	addgene102583	addgene102726
addgene102883	addgene103037	addgene103176	addgene103870	addgene103981	addgene104104
addgene104270	addgene104413	addgene104547	addgene104670	addgene104788	addgene104981
addgene105138	addgene105269	addgene105383	addgene105491	addgene105629	addgene105780
addgene105960	addgene106063	addgene106180	addgene106331	addgene106684	addgene106791
addgene106964	addgene107126	addgene107243	addgene107385	addgene107555	addgene107675
addgene107822	addgene107955	addgene108112	addgene108273	addgene108402	addgene108533
addgene108863	addgene109016	addgene109150	addgene109304	addgene109490	addgene109603
addgene109710	addgene109834	addgene109945	addgene110075	addgene110191	addgene110342
addgene110565	addgene110683	addgene110829	addgene110964	addgene111092	addgene111200
addgene111345	addgene111489	addgene111605	addgene111732	addgene111861	addgene111994
addgene112111	addgene112259	addgene112369	addgene112471	addgene112596	addgene112717
addgene112829	addgene112946	addgene113062	addgene113213	addgene113380	addgene113500
addgene113672	addgene113791	addgene113906	addgene114039	addgene114150	addgene114272
addgene114385	addgene114620	addgene114763	addgene114872	addgene114979	addgene115087
addgene115207	addgene115390	addgene115530	addgene115697	addgene115869	addgene115984
addgene116091	addgene116199	addgene116319	addgene116450	addgene116559	addgene116669
addgene116780	addgene116905	addgene117049	addgene117221	addgene117381	addgene117504
addgene117604	addgene117756	addgene117872	addgene118041	addgene118199	addgene118321
addgene118433	addgene118536	addgene118657	addgene118802	addgene118936	addgene119073
addgene119196	addgene119314	addgene119649	addgene119775	addgene119910	addgene120026
addgene120201	addgene120333	addgene120462	addgene120570	addgene120683	addgene120786
addgene120898	addgene121008	addgene121147	addgene121455	addgene121682	addgene121782
addgene121896	addgene122026	addgene122168	addgene122314	addgene122486	addgene122614
addgene122854	addgene122965	addgene123092	addgene123196	addgene123366	addgene123478
addgene123578	addgene123706	addgene123813	addgene123930	addgene124049	addgene124170
addgene124286	addgene124422	addgene124587	addgene124786	addgene124911	addgene125040
addgene125150	addgene125408	addgene125510	addgene125697	addgene125838	addgene125968
addgene126092	addgene126200	addgene126302	addgene126402	addgene126535	addgene126670
addgene126821	addgene126948	addgene127048	addgene127199	addgene127311	addgene127433
addgene127539	addgene127659	addgene127778	addgene127905	addgene128052	addgene128166
addgene128276	addgene128400	addgene128538	addgene128704	addgene128844	addgene128970
addgene129070	addgene129303	addgene129416	addgene129560	addgene129696	addgene129819
addgene130174	addgene130562	addgene130691	addgene130806	addgene130940	addgene131048
addgene131167	addgene131324	addgene131502	addgene131697	addgene131823	addgene131944
addgene132045	addgene132145	addgene132247	addgene132367	addgene132476	addgene132631
addgene132765	addgene133036	addgene133308	addgene133422	addgene133527	addgene133646
addgene133813	addgene133980	addgene134238	addgene134363	addgene134567	addgene134725
addgene134880	addgene135075	addgene135229	addgene135368	addgene135495	addgene135664
addgene135813	addgene135919	addgene136031	addgene136133	addgene136233	addgene136386
addgene136562	addgene136695	addgene136811	addgene136918	addgene137047	addgene137160
addgene137759	addgene137878	addgene138028	addgene138141	addgene138257	addgene138394
addgene138503	addgene138668	addgene138770	addgene138872	addgene139014	addgene139133
addgene139261	addgene139418	addgene139547	addgene139692	addgene139826	addgene139930
addgene140095	addgene140240	addgene140394	addgene140584	addgene140730	addgene140944
addgene141080	addgene141208	addgene141374	addgene141629	addgene141729	addgene141829
addgene141930	addgene142032	addgene142133	addgene142237	addgene142339	addgene142439
addgene142542	addgene142646	addgene142746	addgene142847	addgene142947	addgene143048
addgene143148	addgene143250	addgene143350	addgene143451	addgene143553	addgene143655
addgene143760	addgene143862	addgene143966	addgene144069	addgene144175	addgene144277
addgene144391	addgene144495	addgene144595	addgene144695	addgene144796	addgene144896
addgene145038	addgene145259	addgene145493	addgene145698	addgene145947	addgene146647
addgene147480	addgene148236	addgene148968	addgene149087	addgene149211	addgene149365
addgene149509	addgene149637	addgene152095	addgene152404	addgene152861	addgene153062
addgene153215	addgene153321	addgene153439	addgene153589	addgene153707	addgene153823
addgene153939	addgene154052	addgene154170	addgene154289	addgene154416	addgene154763
addgene154894	addgene155011	addgene155119	addgene155234	addgene155372	addgene156004
addgene156374	addgene156487	addgene156593	addgene156779	addgene156972	addgene157277
addgene157611	addgene157713	addgene157839	addgene157987	addgene158112	addgene158227
addgene158329	addgene158438	addgene158539	addgene158652	addgene158776	addgene158897
addgene159012	addgene159138	addgene159398	addgene159545	addgene159672	addgene159806
addgene159916	addgene160031	addgene160185	addgene160331	addgene160463	addgene160586
addgene160698	addgene160827	addgene160960	addgene161090	addgene161211	addgene161311

addgene161411	addgene161511	addgene161617	addgene161748	addgene161931	addgene162076
addgene162248	addgene162367	addgene162510	addgene162633	addgene162743	addgene162884
addgene162985	addgene163091	addgene163273	addgene163380	addgene163519	addgene163643
addgene163759	addgene163912	addgene164033	addgene164195	addgene164449	addgene164579
addgene164699	addgene164838	addgene164955	addgene165068	addgene165419	addgene165532
addgene165798	addgene165970	addgene166109	addgene166210	addgene166488	addgene166742
addgene166851	addgene166969	addgene167196	addgene167331	addgene167556	addgene167669
addgene167831	addgene167948	addgene168066	addgene168198	addgene168330	addgene168430
addgene168558	addgene168849	addgene168960	addgene169111	addgene169227	addgene169360
addgene169468	addgene169594	addgene169741	addgene169914	addgene170036	addgene170177
addgene170325	addgene170449	addgene170601	addgene170729	addgene170860	addgene170960
addgene171084	addgene171225	addgene171335	addgene171509	addgene171654	addgene171788
addgene171941	addgene172136	addgene172309	addgene172427	addgene172550	addgene172669
addgene172838	addgene172960	addgene173098	addgene173212	addgene173316	addgene173416
addgene173541	addgene173645	addgene173784	addgene173935	addgene174042	addgene174177
addgene174300	addgene174429	addgene174546	addgene174663	addgene174797	addgene174920
addgene175026	addgene175140	addgene175266	addgene175419	addgene175532	addgene175783
addgene175919	addgene176068	addgene176177	addgene176297	addgene176435	addgene176562
addgene176701	addgene176802	addgene176918	addgene177034	addgene177141	addgene177258
addgene177420	addgene177723	addgene177850	addgene177969	addgene178108	addgene178223
addgene178336	addgene178465	addgene178669	addgene178773	addgene178915	addgene179035
addgene179157	addgene179369	addgene179511	addgene179668	addgene179811	addgene179963
addgene180193	addgene180339	addgene180450	addgene180561	addgene180674	addgene180792
addgene181772	addgene181900	addgene182027	addgene182206	addgene182350	addgene182494
addgene182617	addgene182818	addgene182951	addgene183129	addgene183270	addgene183385
addgene183526	addgene183663	addgene183801	addgene183922	addgene184044	addgene184160
addgene184393	addgene184531	addgene184667	addgene184797	addgene184908	addgene185019
addgene185152	addgene185253	addgene185356	addgene185500	addgene185640	addgene185771
addgene185907	addgene186067	addgene186183	addgene186361	addgene186475	addgene186644
addgene186777	addgene186893	addgene187063	addgene187170	addgene187330	addgene187482
addgene187715	addgene187846	addgene187976	addgene188084	addgene188273	addgene188494
addgene188631	addgene188759	addgene188860	addgene188994	addgene189719	addgene189859
addgene189975	addgene190122	addgene190249	addgene190433	addgene190668	addgene190833
addgene190980	addgene191083	addgene191202	addgene191333	addgene191480	addgene191601
addgene191720	addgene191855	addgene192057	addgene192210	addgene192329	addgene192430
addgene192572	addgene192725	addgene192852	addgene192981	addgene193097	addgene193206
addgene193314	addgene193423	addgene193574	addgene193681	addgene193809	addgene193976
addgene194138	addgene194268	addgene194385	addgene194512	addgene194631	addgene194809
addgene194973	addgene195103	addgene195240	addgene195402	addgene195516	addgene195658
addgene195764	addgene195875	addgene195980	addgene196145	addgene196267	addgene196490
addgene196659	addgene196777	addgene196878	addgene197012	addgene197160	addgene197261
addgene197401	addgene197537	addgene197675	addgene197775	addgene197883	addgene198023
addgene198183	addgene198339	addgene198480	addgene198610	addgene198740	addgene198850
addgene198986	addgene199088	addgene199214	addgene199352	addgene199495	addgene199680
addgene199942	addgene200063	addgene200187	addgene200299	addgene200407	addgene200527
addgene200793	addgene200935	addgene201098	addgene201249	addgene201402	addgene201517
addgene201640	addgene201870	addgene201987	addgene202107	addgene202213	addgene202367
addgene202519	addgene202669	addgene202818	addgene202922	addgene203032	addgene203137
addgene203285	addgene203423	addgene203552	addgene203676	addgene203806	addgene203947
addgene204071	addgene204183	addgene204309	addgene204479	addgene204588	addgene204749
addgene204958	addgene205065	addgene205186	addgene205330	addgene205451	addgene205566
addgene205679	addgene205795	addgene205895	addgene205996	addgene206145	addgene206273
addgene206418	addgene206823	addgene207005	addgene207131	addgene207253	addgene207358
addgene207487	addgene207606	addgene207726	addgene207843	addgene207949	addgene208105
addgene208279	addgene208405	addgene208630	addgene208739	addgene208866	addgene208993
addgene209101	addgene209253	addgene209389	addgene209510	addgene209699	addgene209847
addgene209970	addgene210179	addgene210314	addgene210467	addgene210626	addgene210762
addgene210877	addgene210977	addgene211077	addgene211177	addgene211277	addgene211380
addgene211518	addgene211636	addgene211769	addgene211894	addgene212050	addgene212173
addgene212401	addgene212507	addgene212665	addgene212766	addgene212873	addgene212991
addgene213178	addgene213278	addgene213378	addgene213559	addgene213752	addgene213908
addgene214053	addgene214214	addgene214326	addgene214439	addgene214543	addgene214648
addgene214789	addgene214907	addgene215054	addgene215159	addgene215281	addgene215427

addgene215544	addgene215646	addgene215821	addgene215923	addgene216026	addgene216130
addgene216256	addgene216414	addgene216524	addgene216672	addgene216837	addgene216953
addgene217073	addgene217187	addgene217299	addgene217439	addgene217615	addgene217736
addgene217894	addgene218126	addgene218250	addgene218427	addgene218606	addgene218716
addgene218864	addgene218980	addgene219081	addgene219181	addgene219281	addgene219381
addgene219560	addgene219750	addgene219889	addgene220068	addgene220193	addgene220347
addgene220552	addgene220666	addgene220776	addgene220900	addgene221062	addgene221197
addgene221312	addgene221447	addgene221565	addgene221711	addgene221853	addgene221981
addgene222116	addgene222336	addgene222491	addgene222603	addgene222745	addgene222934
addgene223042	addgene223147	addgene223327	addgene223442	addgene223560	addgene223679
addgene223791	addgene223923	addgene224034	addgene224179	addgene224348	addgene224477
addgene224687	addgene224843	addgene225135	addgene225299	addgene225479	addgene225649
addgene225780	addgene225949	addgene226162	addgene226306	addgene226452	addgene226598
addgene226733	addgene226906	addgene227123	addgene227249	addgene227354	addgene227489
addgene227598	addgene227756	addgene227901	addgene228014	addgene228147	addgene228262
addgene228386	addgene228554	addgene228661	addgene228797	addgene228938	addgene229174
addgene229278	addgene229378	addgene229486	addgene229604	addgene229754	addgene229858
addgene229968	addgene230130	addgene230240	addgene230345	addgene230447	addgene230551
addgene230655	addgene230758	addgene230868	addgene231035	addgene231202	addgene231316
addgene231514	addgene231635	addgene231770	addgene231924	addgene232096	addgene232236
addgene232378	addgene232496	addgene232652	addgene232843	addgene232970	addgene233102
addgene233226	addgene233361	addgene233462	addgene233598	addgene233812	addgene233929
addgene234038	addgene234156	addgene234256	addgene234393	addgene234555	addgene234684
addgene234827	addgene235045	addgene235165	addgene235316	addgene235536	addgene235676
addgene235804	addgene235905	addgene236017	addgene236147	addgene236272	addgene236439
addgene236547	addgene236710	addgene236832	addgene236965	addgene237083	addgene237215
addgene237363	addgene237525	addgene237645	addgene237877	addgene237987	addgene238181
addgene238332	addgene238513	addgene238642	addgene238759	addgene238871	addgene239051
addgene239236	addgene239385	addgene239530	addgene239648	addgene239782	addgene239907
addgene240039	addgene240178	addgene240322	addgene240550	addgene240681	addgene240983
addgene241110	addgene241255	addgene241392	addgene241723	addgene242024	addgene242179
addgene242461	addgene242599	addgene242753	addgene242882	addgene243050	addgene243740
addgene244087	addgene244261	addgene244501	addgene244603	addgene244703	addgene244803
addgene244911	addgene245063	addgene245298	addgene245556	addgene245666	addgene245858
addgene246044	addgene246282	addgene246399	addgene246642	addgene246815	addgene247073
addgene247293	addgene247422	addgene247646	addgene247999	addgene248199	addgene248731
addgene248876	addgene249268				